

SAFWAN ALI

AI Engineer • Data Scientist • LLM & Full-Stack AI Developer

safwanalimukaddam@gmail.com • +92 316 258 7066 • github.com/Safwan2003 • linkedin.com/in/safwan-ali-281aa1275
• Karachi, Pakistan

PROFESSIONAL SUMMARY

AI Engineer and final-year Computer Science graduate (Salim Habib University, June 2026) with nearly a year of professional experience building production-grade AI systems at MarkyTech. Hands-on expertise across the full AI development lifecycle — from LLM-powered chatbots and RAG pipelines to ML model development and full-stack AI web applications. Delivered a SUPARCO-aligned Final Year Project combining satellite imagery, deep learning, and LLMs for lunar soil analysis. Adept at translating complex business requirements into scalable, deployable AI solutions using modern frameworks including LangChain, AutoGen, FastAPI, and Next.js.

PROFESSIONAL EXPERIENCE

AI Engineer

Aug 2024 – Present

MarkyTech • Karachi, Pakistan

Promoted from AI Intern to AI Engineer within the same engagement. MarkyTech is a digital solutions agency specializing in social media marketing, web development, and AI-powered products.

- Engineered the lead generation component for Busman AI, an AI-powered business automation product, integrating conversational AI to qualify and capture leads via intelligent chat flows.
- Built and deployed LLM-powered chatbots for clients using OpenAI GPT APIs, Gemini API, and Claude API, incorporating LangChain and LlamaIndex for context management and retrieval-augmented generation (RAG) pipelines.
- Developed multi-agent AI workflows using AutoGen and AgentScope frameworks to automate complex, multi-step business processes with minimal human intervention.
- Implemented web scraping and data ingestion pipelines using Crawl4AI to feed structured marketing and business intelligence data into AI systems.
- Built production REST APIs with FastAPI serving AI model inference endpoints, and developed frontend interfaces using Next.js to create end-to-end AI-powered web applications.
- Automated internal Python scripting tasks for data processing, report generation, and workflow orchestration across the engineering team.

EDUCATION

Bachelor of Science in Computer Science

2022 – June 2026

Salim Habib University — Karachi, Pakistan — Specialization: Data Science, Machine Learning & AI

- Final Year Project: AI-Driven Lunar Soil Composition Analysis System — developed in alignment with SUPARCO's ICUBE-Qamar lunar satellite mission.
- Relevant coursework: Machine Learning, Deep Learning, Database Systems, Statistics & Probability, Data Structures & Algorithms.

KEY PROJECTS

AI-Driven Lunar Soil Composition Analysis (FYP — SUPARCO) *Python • SAM 2.1 • ResNet-18 • Gemini 2.0 Flash • Streamlit*

- Built an end-to-end pipeline analyzing 457 real Chang'e 3 Yutu rover images to estimate 6 lunar mineral oxides (FeO, TiO₂, MgO, SiO₂, Al₂O₃, CaO), calibrated to Chang'e 3 APXS ground-truth data and aligned with Pakistan's ICUBE-Qamar satellite sensor specifications.

- Integrated SAM 2.1 for terrain segmentation and ResNet-18 deep learning classifier to identify geological units; automated scientific report generation via Gemini 2.0 Flash LLM.
- Delivered a Streamlit Mission Control dashboard with real-time telemetry and physics-based geochemical constraints (oxide-sum conservation).

[GitHub](#)

Heart Disease Prediction — Random Forest Classifier

Python · Scikit-learn · Streamlit (Deployed)

- Developed and deployed a Random Forest classifier for clinical heart disease risk prediction with feature selection, data preprocessing, and hyperparameter tuning.
- Shipped as a live Streamlit Cloud application serving real-time inference.

[GitHub](#) | [Live Demo](#)

Fruit Classification CNN

Python · TensorFlow · Keras

- Built a CNN with data augmentation and batch normalization for multi-class fruit image classification; applied transfer learning and hyperparameter tuning to maximize validation accuracy.

[GitHub](#)

Customer Segmentation — K-Means Clustering

Python · Scikit-learn · Matplotlib

- Implemented K-Means clustering on customer purchasing data with elbow-method optimization, producing actionable marketing segment visualizations.

[GitHub](#)

TECHNICAL SKILLS

Languages: Python, JavaScript

LLM & AI Frameworks: LangChain, LlamaIndex, AutoGen, AgentScope, OpenAI API, Gemini API, Claude API, RAG Pipelines

ML / Deep Learning: TensorFlow, Keras, Scikit-learn, XGBoost, CNNs, SVMs, SAM 2.1, ResNet

Data & Scraping: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Crawl4AI

Backend & APIs: FastAPI, Node.js, Express.js

Frontend: Next.js, React, Streamlit

Databases: PostgreSQL, MongoDB

DevOps & Tools: Docker, Git, GitHub, Jupyter Notebook, VS Code, Vercel

Languages (Human): English (Professional), Urdu (Native)

ACHIEVEMENTS & NOTABLE WORK

- Promoted from AI Intern to AI Engineer at MarkyTech within the same engagement, demonstrating rapid professional growth in a production environment.
- Delivered a core lead generation component for Busman AI, a commercial AI-powered SaaS product, while still a final-year student.
- SUPARCO FYP alignment — Final Year Project designed for direct application to Pakistan's ICUBE-Qamar lunar satellite mission, one of Pakistan's most significant space programs.
- 24 public GitHub repositories spanning deep learning, NLP, agentic AI, clustering, and full-stack AI applications — github.com/Safwan2003.

References available on request • Portfolio: github.com/Safwan2003